

1.1 Problem Statement

Problem

Let Σ be a finite alphabet and fix a word length L . You are given:

- a **start word** $s \in \Sigma^L$,
- a **target word** $t \in \Sigma^L$,
- a **dictionary** $D \subseteq \Sigma^L$ of valid words.

A *valid transformation* changes exactly one letter of a word, yielding another word in D . A transformation chain is a sequence

$$s = w_0 \rightarrow w_1 \rightarrow \cdots \rightarrow w_k = t$$

where each $w_i \in D \cup \{s\}$ and consecutive words differ in exactly one position.

Find the minimum chain length $k + 1$. Return 0 if no chain exists.

1.2 Running Example

Parameter	Value
Start s	<code>hit</code>
Target t	<code>cog</code>
Dictionary D	<code>{hot, dot, dog, lot, log, cog}</code>

1.3 Questions

- Q1.** Explain why this problem reduces to a graph shortest-path problem and describe the graph model.
- Q2.** Justify why **BFS** is the appropriate algorithm (not DFS, not Dijkstra).
- Q3.** Show the complete BFS layers starting from $s = \text{hit}$.
- Q4.** Compute the minimum chain length for this example.
- Q5.** Suppose $|D| = n$ and each word has length L . Derive the time complexity.
- Q6.** Write pseudocode for the BFS-based solution for general inputs.

